

# DAMATH DE SILVA



BSC ENGINEERING HONOURS  
COMPUTER ENGINEERING UNDERGRADUATE,  
UNIVERSITY OF JAFFNA.

---

 [GitHub](#)  [LinkedIn](#)  [Portfolio](#)  [damathdesilva@gmail.com](mailto:damathdesilva@gmail.com)  Negombo, Sri Lanka  +94773419799

---

## PROFILE

---

I am a final-year Computer Engineering undergraduate at the University of Jaffna, with a strong passion for software engineering, AI/ML, full-stack development, and embedded systems. My interests lie in solving real-world problems by integrating intelligent systems with scalable software solutions.

Proficient in **Java**, **Python**, **C++**, **React.js**, and **Express.js**, I have hands-on experience building responsive web apps, **RESTful APIs**, and working with **MySQL** and **MongoDB**. I also work on embedded systems using Arduino and ESP32, merging hardware and software for real-time, intelligent applications.

I am a fast learner and a collaborative team player. I enjoy exploring new frameworks and contributing to tech-driven innovation. I am currently seeking an internship opportunity where I can apply my skills, expand my knowledge, and contribute meaningfully to efficacious projects.

## PROJECTS

---

- 1. SHOPLANE (E-commerce platform)**   Jun 2025 – Present  
**Technologies:** MongoDB, Express.js, React.js, Node.js, JWT, Tailwind CSS, React Icons, Postman
  - Developed a modern, scalable e-commerce platform using the MERN stack, featuring secure user authentication (JWT), role-based access, and robust RESTful APIs with Express.js and MongoDB.
  - Designed a responsive UI with React.js, Tailwind CSS, and React Icons, ensuring dynamic interactions, while testing backend reliability through Postma.
- 2. SMART WATER – IoT-Based Household Water Management System (Group Project)**  Jan 2025 - May 2025  
**Technologies:** ESP32, C, React, Firebase  
**My role:** Focused on embedded C programming, sensor data processing, and cloud integration
  - Developed firmware in C for an ESP32-based system to monitor and report household water inflow/outflow, level, and quality (Total Dissolved Solids).
  - Integrated multiple sensors and enabled secure data transmission to Firebase.
- 3. BOOKIFY – Booking and Management Platform (Group Project)**  Jan 2025 - May 2025  
**Technologies:** Spring Boot, React.js, MySQL, JWT, Git & GitHub, Postman (API testing)  
**My role:** Project Manager and Backend Developer  
**Tools:** Jira, Figma
  - Built a web-based platform for managing one-time and recurring bookings with features like automated reminders, client profiles, feedback, and analytics.
- 4. Student Attendance Management System (2 Members Project)**  Mar 2024 - Oct 2024  
**Technologies:** HTML, CSS, JS, PHP, MySQL

- Built a role-based system for managing course-wise student attendance, allowing students to mark attendance and manage subject enrollments, with lecturer approval and tracking functionalities

## 5. Optimizing AI Algorithms for Low-Power and Edge Devices in Resource-Constrained Environments (Final Year Research Project) Jan 2025 - Present

- Developed an agent-based Edge AI system for dynamic selection and optimization of pretrained models using metadata (modality, RAM, CPU, power, accuracy).
- Enabled real-time, multimodal inference (image, audio, text, sensor) on microcontrollers with lightweight deployment strategies for resource-constrained environments.

## EDUCATION

---

**BSc Engineering (Hons) in Computer Engineering** Apr 2022 - Present

Current GPA: 3.41

**GCE Advanced Level Examination – Physical Science Stream** 2020

Maris Stella College

Combined Mathematics – B, Physics – B, Chemistry – B, General English - A

## SKILLS & ABILITIES

---

- **Programming Languages:** Python, C++, Java, PHP, JavaScript
- **Web & Software Development:** React.js, Express.js, HTML, CSS, REST APIs, SQL
- **Databases:** MySQL, MongoDB
- **Machine Learning & AI:** Scikit-learn, TensorFlow, OpenCV
- **Design Tool:** Figma
- **Project Management:** Jira
- **Version Control:** Git, GitHub

## VOLUNTEERING

---

Secretary, IEEE Robotics and Automation Society (RAS) - University of Jaffna Feb 2024-Present

## AWARDS & ACHIEVEMENTS

---

- YARL XTREME (1<sup>st</sup> place)  Oct 2024
- IEEE Xtreme 18.0 (Sri Lanka Rank: 116)  Oct 2024
- MoraXtreme 9.0 (61<sup>st</sup> out of 300+ teams)  Oct 2024
- CodeRally 5.0 (12<sup>th</sup> place out of 40 teams)  Sep 2024

## NON-RELATED REFEREES

---

**DR. (Mrs.) J. Jananie**

Senior Lecturer Grade II

Department of Computer Engineering,

Faculty of Engineering,

University of Jaffna

[jananie@eng.jfn.ac.lk](mailto:jananie@eng.jfn.ac.lk)

+94 70 490 4957

**Mr. Y. Pirunthapan**

Lecturer

Department of Computer Engineering,

Faculty of Engineering,

University of Jaffna

[pirunthapan@eng.jfn.ac.lk](mailto:pirunthapan@eng.jfn.ac.lk)

+94 21 206 0160